

Brain Tumor MRI Image Segmentation and Classification based on Deep Learning Techniques

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Abstract—Detection and diagnosis of brain tumors is important for improving the possibility of successful treatment and recovering. Magnetic resonance imaging (MRI) is widely used imaging method for treating and recovering brain tumors. However, manual identification of brain tumors from a large amount of MRI images is time-consuming and requires specialized expertise. To overcome these challenges, computer-assisted intelligent systems are increasingly being used to speed up the medical assessment as well as treatment recommendations. The aim of our research is for coming up with a deep learning system that can segment and classify tumors in brain. The U-Net model is used for segmentation of the MRI images, while Convolution Neural Network (CNN) is used for the classifying brain tumors. Performance metrics such as accuracy, precision and recall are used to evaluate the effectiveness of this approach. The suggested CNN classifier has given the accuracy of nearly 98% for both the training and validation data. By using deep learning techniques, the following system attempts to provide accurate and effective segmenting and classifying tumors.

Keywords— Brain tumor, segmentation, classification, U-Net, convolution neural network (CNN), magnetic resonance imaging (MRI)

I. INTRODUCTION

Brain tumors are characterized from more than necessary developing brain tissues. We may not know what cause tumors in brain or what symptoms they can cause. As a result, some people may develop brain tumors without realizing how terrible their situation is. Tumors are categorized into three categories depending on the cells that give birth to them: glioma, pituitary, and meningioma. Meningiomas. Meninges or membranes that cover and protect brain is where Meningiomas grow. The brain is the most visible component of the central nervous system's as well as peripheral nervous systems. Non-invasive investigations into the composition and function of the brain are required by both medical practitioners and scientists. A bulge in skull that appears due to excessive cell division is called a brain tumor [13]. Brain cells can die by brain tumor in both direct and indirect ways, like by inducing irritation, pressing against nearby brain tissue thus an edoema, or by putting stress when they develop. Two types of tumors are referred as benign (cancerous) or malignant (non cancerous). Cancerous tumors are the pituitary tumors that develop inside the pituitary glands. Doctors and radiologist can make better decisions and increase patient survival rate if brain tumor can be detected in early stages. MRI is very important diagnostic tool that helps in early detection, therapy and observing of brain tumors. However, interpreting MRI images is a complex and time-consuming task that requires specialized expertise. Additionally, manual

detection of malignancies can be challenging and may vary among different clinicians due to their experience. Effective segmentation and classification of MRI images can also be difficult. Objective here is to make an intelligent network that can accurately detect malignant cells in brain MRI images. Various researchers from different fields have utilized image recognition systems to detect brain tumor cells over time. The system utilizes a range of algorithms to effectively segment and classify the images, allowing for more precise diagnosis and treatment of brain tumors. The ultimate aim is improving patient outcomes and reduces the need for invasive procedures.

Aim is to develop an effective model to facilitate automated tumor detection. The conventional approach to classifying brain tumors involves segmenting tumors based on regions prior to feature extraction and sorting. The proposed method involves using a U Net neural network to segment brain tumors from MRI images, followed by using a convolution neural network architecture to classify the resulting tumor-containing images into glioma, meningioma, and pituitary tumor categories. U-Net and VGG-16 was used here for brain tumor segmentation and classification. U-Net is a popular convolutional neural network architecture used for medical image segmentation, while VGG-16 is a deep convolutional neural network architecture used for image classification. For brain tumor segmentation, U-Net model was used to predict tumor regions in brain images.

The accuracy of the segmentation and classification models was evaluated using metrics such as Dice coefficient and classification accuracy. Our experiments showed that both the U-Net and VGG-16 models performed well for brain tumor segmentation and classification, achieving high accuracy rates.

Overall, the combination of U-Net and VGG-16 models allowed for accurate and efficient segmentation and classification of brain tumors, which could potentially aid medical professionals in the accurate diagnosis and treatment of brain tumors.

II. RELATED WORK

Arpit Mohan; Deependra Rastogi - 2022 [1], The authors of this paper utilized magnetic resonance (MR) images of the brain to segment brain tissues into different categories including normal tissues (white matter, grey matter, and cerebrospinal fluid) as well as tumor-infected tissues. Pre-processing techniques were employed to enhance the signal-to-noise ratio and eliminate unwanted noise. To improve the skull stripping performance, a threshold-based skull stripping algorithm was employed. The second paper was by Prashant

Johri; Varun Tiwari – 2021 [2], The authors conducted a survey to examine the different techniques implemented in Medical Image Processing for identifying tumors in brain from MRI pictures. So, this paper summarizes the findings of the survey and provides a comprehensive list of the various techniques currently in use, along with a brief description of each. Among all the steps involved in the process of detecting tumors, segmentation is considered to be the most critical. Hayder Abdullahi; Mohd Zubir Mat – 2014 [3], suggested that the identification of brain tumors through MRI data might be divided into four modules: pre processing, segmentating image, extracting features, and classifying image. A survey is proposed to cover these stages comprehensively.

K. M. Ahmmed; Misbah Ul – 2019 [4], They have done some of the recent research work on detecting and segmenting brain tumor is reviewed. Different techniques used by scientists to identify tumor tissues from the MRI data is illustrated. With this review it was discovered that detection of Brain tumors with help of machines is the most researched areas. Daisuke Hirahara – 2019 [5], Proposed a new method for taking a look at identifying brain tumors through MRI images involving a hybrid technique using SVM and fuzzy c-means. The algorithm includes pre-processing steps like contrast enhancement and stretching to mid-range, as well as morphological operations and double thresholding for skull stripping. Fuzzy c-means is implemented in segmentation, and the GLRLM, ie grey level run length matrix is implemented for extracting features. Finally, Linear, Quadratic, and Polynomial SVM techniques are applied to classify MRI data as either 'non-tumor' or 'tumor'. The algorithm was tested on a real 120 patients' MRI brain images, with 96 images used for training the SVM classifier and 24 images used for testing. The SVM classifier achieved 91.66% accuracy with the Linear kernel function, 83.33% accuracy with the Quadratic kernel function, and 87.50% accuracy with the Polynomial kernel function, with 100% specificity.

In Gökay Karayeğen; Mehmet Feyzi Akşahin – 2021 [6], The MRI brain image was pre-processed using the discrete wavelet transform (DWT) technique, which involves decomposing the image into different frequency bands. Thresholding was applied to the wavelet coefficients to denoise the image. Next, a genetic algorithm was used to identify pixels that correspond to brain tumors. This algorithm searched for the best combination of features extracted by a criterion and used k-means clustering to guide the search for optimal or suboptimal data partitions. The accuracy of the segmentation achieved with this method ranged from 82% to 97% of the detected brain tumor pixels, as compared to the ground truth. However, problem with approach here is because the wavelet transform needs large amount of storage and is computationally expensive. However, in T. A. Jemimma; Y. Jacob Vetharaj – 2018 [7], The methodology employed in their study consists of several stages, including preprocessing, segmentating, extracting features, and classifying. The first stage utilizes a morphology technique that employs double thresholding to get rid of the skull from the images. Current research compares two distinct techniques for identifying tumors in MRI images.

Aditya Raut; Jeevan Bhagade; Jyoti; Sachin - 2020 [8], Methodology consists of stages such as histogram analysis, re-sampling, k-NN algorithm, and distance matrix. Initially, histogram analysis is used to determine the pixel value distribution in the MRI image. Re-sampling is performed to

resize the image to 629 x 839, which helps in improving its geometrical representation. The k-NN algorithm is used for brain tumor classification, where training is done with a k-value. Manhattan metric is used to calculate the distance of the classifier. The algorithm implementation is performed using LabView. The proposed algorithm is tested on 48 MRI images, resulting in an identification score of approximately 95%. Annisa Wulandari; Riyanto Sigit; Mochamad MobedBachtiar – 2018 [9], They used Naïve Bayes and SVM Classifier approach which gave them the accuracy 87.23% and 91.49% respectively. SVM give better classification accuracy. For implementation MATLAB is used. Method here has limitations like not finding out the with precision the boundary of affected area. Neslihan Kurat; Necla Özkaya – 2014 [10] paper was a novel approach of an effective automatic classification method for brain MRI is projected implementing Adaboost machine learning. It gives 89.90% accuracy and result in normal brain or in Malignant or Benign type of tumor.

III. PROPOSED WORK

This section, aim is to explain proposed framework used in this study. The methodology consists of three primary steps, namely preprocessing, segmentation, and classification.

The first step of preprocessing involves identifying the regions of interest (ROI) by performing skull stripping. Skull stripping is an essential process in the field of biological and medical image processing as it plays a vital role in accurately identifying tumor tissues from MRI images. The skull stripping process involves the removal of non-brain tissues such as skin, skull, and other unwanted elements from the brain MRI images. The outcome of this process is a clean and noise-free MRI image that can be used for further analysis.

In the second step, tumor segmentation is performed using the U-Net model. This model is a convolutional neural network that is usually implemented in segmentation of images. U-Net model is trained on a dataset of brain MRI images with pre-labeled tumor regions. The model learns the features of the tumor regions from these images and generates a binary mask that highlights the tumor region in the input image.

Finally, tumor classification is performed by implementing a CNN. Following model is trained on the segmented images from the previous step and can classify the tumors into three categories: meningioma, glioma, and pituitary. Network is trained to recognize specific patterns and features in the MRI images that are indicative of each tumor type. The model can accurately classify the tumors based on these features and provide insights into the type and severity of the brain tumor.

In conclusion, the proposed framework in this study provides an efficient and effective approach for detecting and classifying brain tumors from MRI images. The combination of skull stripping, U-Net model, and convolutional neural network has shown promising results in accurately identifying and classifying brain tumors into specific categories.

For brain MRI image classification and segmentation, this model employs the U-Net neural network and convolution neural network. U-Net aids in the segmenting tumor in brain MRI images, whereas CNN aids in the classification of tumors. Below diagram illustrates the workflow of a segmenting and classifying tumor in input images. This paper

describes how to utilize neural network models to segment the tumor part and classify it.

First the collected images are preprocessed in which blurred and corrupted images are filtered and it also include removing of noise from the interior part. In order to create an expert system that can accurately detect brain cancers at an early stage, the framework in this study included the installation of a U-Net model for segmentation and CNN to classify MRI images.

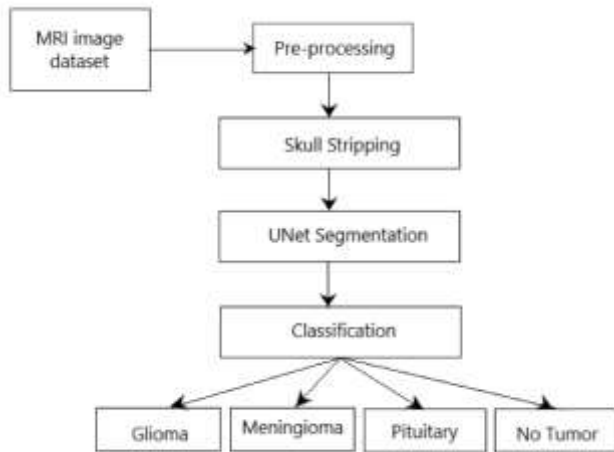


Fig. 1. Architecture diagram of segmentation and classification of brain tumor

IV. IMPLEMENTATION

The entire project is developed and maintained using colab. Each module was divided into distinct files with their own class names to guarantee that the team could operate effectively and quickly.

The project was carried out in the subsequent stages-

A. Data Collection

Data collection involves acquiring and analysing data gathered from an extensive range of sources, in order to respond to specified research problems, test hypotheses, and assess results, a systematic approach has been devised. In this project have used brain tumor datasets of segmentation and classification which is available Kaggle. Brain tumor segmentation data set contains around 3064 images of brain tumor as well as the mask of each and every image. The Brain tumor Classification (MRI) data set serves as the basis for the classification model. This dataset includes 3264 files with the labels "glioma," "meningioma," "pituitary," and "no tumor images," and it is split into testing and training set.

B. Pre-Processing

At this stage, the regions of interest (ROI) are retrieved utilizing skull stripping. Moreover, pre-processing helps to enhance several MRI images metrics, which includes signal-to-noise ratio, by removing extraneous ambient noise and bits and smoothing the interior portion of the region while maintaining its boundaries. In this procedure, the greatest intensity values included in a picture are divided by each pixel's intensity. Binary quantization may be produced through normalization by expanding the source. These MRI images can assist in preventing classifications from being impacted by changes in grayscale value.

Skull stripping refers to the removal of non-brain structure and unwanted portions of image from scanned image to have the required image for tumor detection. Scanned image consists of brain area, scalp, skull and dura. The unwanted portions can be separate with the aid of rim of cerebrospinal fluid (CSF). Skull removing can be done with the help of intensity thresholding followed by morphological operation to obtain required brain area for tumor detection.

The method is such that it satisfies two conditions. One is that the connection between brain and non brain tissues should be weak. Second is that the mask produced by intensity thresholding should preserve as much brain as possible. The selection of threshold value is critical in this scenario because, non desirable inclusion of dura might be included if threshold values are too low. Too high threshold values can provide a clear separation of brain and non-brain structures although image suffers some brain erosion.

C. Segmentation

In this research, the U-Net model was utilized tumor segmentation, a popular architecture for biomedical image segmentation. The U-Net model is named as such because of its U-shaped structure, contraction and expansion takes place. The path for contraction is composed of down-sampling or down-convolution layers, while the expanding path includes up-sampling or up-convolution layers. This model is particularly advantageous when dealing with images because of its ability to efficiently partition pictures into semantically related regions. The U-Net design only contains spiral coating stacks and max-combining tier stacks, lacking a sufficiently related feedforward layer or thick coating. This model was initially planned for 572 representations, but can be surely used to operate accompanying some image breadth by stacking diversified convolution tiers. The U-Net model has happened widely utilized for biomedical figure separation and has delivered superior effects in tracking containers. It is persuasive when used accompanying many instances and can create accurate consequences.

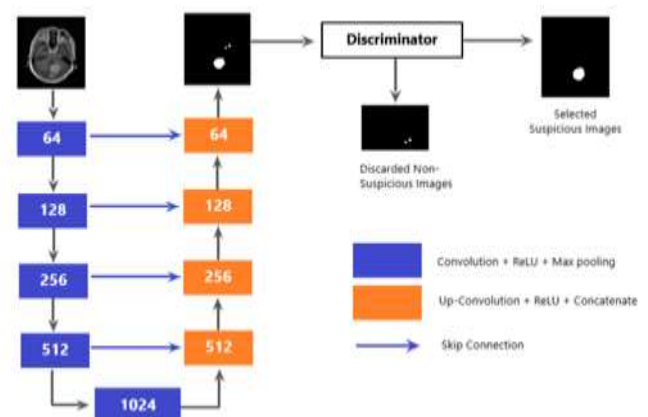


Fig. 2. Proposed lightweight implementation of U-net architecture

Suspicious images in brain segmentation using U-Net contain regions likely to be brain tumors, while non-suspicious images do not. The U-Net model segments brain images into different regions, assigning probabilities of tumor or normal tissue.

D. Classification

This study, has proposed a CNN model for classifying brain tumors into different categories: meningioma, glioma, pituitary as well as no tumor. The proposed architecture consists of 16 layers that help the classifier to effectively categorize the brain tumor images. To train and test the model, the researchers collected a dataset from Kaggle which contained 3264 photos in each of the four classes.

CNN, a deep learning design which is commonly used for tasks such as image recognition, object classification, and face recognition [11]. It works by taking an input image and assigning weights to the features in the image to differentiate one image from another. The architecture of a typical CNN includes convolution layers with filters, layers of pooling, and layers which are fully connected. The convolution layers use filters for extracting input images features, while the pooling will decrease the spatial area of the feature maps in output. Finally, the layers that a fully connected helps to classify the output features into the desired categories.

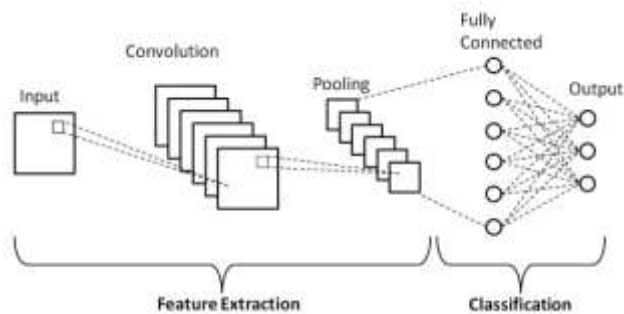


Fig. 3. Basic CNN Architecture

Let's take a closer look at the four primary layers of CNN:

The Convolution Layer is a crucial element of CNN, as it extracts features from the input image. This layer applies a set of filters to the input image, resulting in the creation of feature maps. These feature maps highlight a range of patterns, including edges, corners, and textures, within the image. Each filter is a small matrix of weights that convolves with the input image to produce a feature map. Spatial dimensions are reduced by pooling. It down samples the feature maps, this layer makes them more computationally efficient to process. Different kinds of pooling layers, such as average and max pooling. To select max value within a specific region of the feature map Max pooling is used, while average values are calculated by average pooling. The Fully Connected Layer is responsible for classifying the input image based on the features that were extracted. This layer takes the output from the previous layers and passes it through a set of neurons. Each neuron computes a weighted sum of its inputs and applies an activation function. Fully connected layer's output is a set of class scores that indicate the chances of which class input MRI image belongs to.

Output Layer is final layer of CNN. Its primary function is to produce prediction of input image belonging to a particular class. Forms of output layer varies depending on the problem needing classification. In binary classification type, the layer may consist of a single neuron with a sigmoid activation function. In multi-class problems, the layer may consist of multiple neurons with a SoftMax activation function. Visual Geometry Group (VGG-16) is a convolution neural network (CNN) architecture that was first presented

from Visual Geometry Group (VGG) at the Oxford University. VGG-16 is a relatively simple and straightforward CNN architecture that has become a popular choice for various image recognition tasks.

The VGG-16 architecture has sixteen layers, thirteen convolution layers and three layers that are fully connected. The convolution layers are arranged in sequential groups of two or three layers, with each group having a max pooling. The max-pooling layers help to down-sample the feature maps and reduce the spatial dimensionality, making it easier for the model to learn abstract features. The VGG16 architecture can also be used for differentiating tumors into four categories. To do this VGG16 network is first trained on a large dataset of brain tumor images labeled with their respective categories [12]. The network learns to extract important features from the images and classify them into the four categories. Once the network is trained, it can be used to classify new brain tumor images into one of the four categories. The image is first preprocessed to match the input size of the VGG16 network (224 x 224 pixels). Then, the image is passed through the network, and the output is a probability distribution over the four categories. The category with the highest probability is chosen as the classification result. This classification can help doctors make more informed decisions about treatment options and improve patient outcomes.

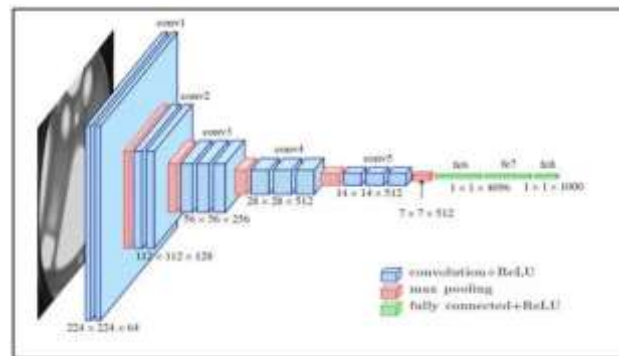


Fig. 4. A Standard VGG 16 Network Architecture

V. RESULTS AND DISCUSSION

The study aimed to develop an accurate and efficient system for brain tumor segmentation and classification. To achieve this, the researchers used a Kaggle dataset containing brain tumor images and masks for segmentation. They employed a popular segmentation model, U Net, to predict the tumor regions in the images. The U Net model is a convolution neural network architecture that has shown great success in medical image segmentation tasks. The predicted tumor regions were then passed on to the classification model for further analysis.

For the classification task, the dataset used contained four types of brain tumors. A CNN model was trained on this dataset to classify the brain tumors into these four categories. CNN, a neural network that particularly well-suited with image classification tasks. It consists of many CNN layers that can do feature extraction from input images, followed by fully connected layers that perform the classification. The accuracy of the classification model was reported to be around 90%, indicating that the model performed well in distinguishing between the different types of brain tumors. The use of CT scan images as input to the CNN model

allowed for efficient and accurate classification of brain tumors, with the added benefit of being non-invasive.

In summary, the study demonstrates the effectiveness of using a multi-layer CNN model for both brain tumor segmentation and classification tasks. The high accuracy achieved by the system could potentially aid medical professionals in the accurate diagnosis and treatment of brain tumors, ultimately leading to better patient outcomes.

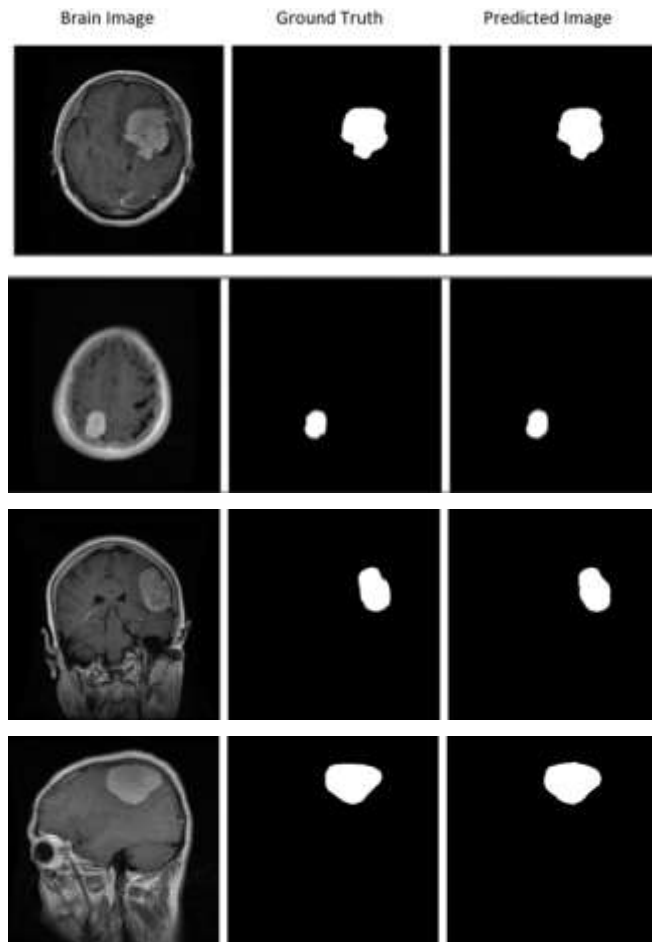


Fig. 5. Result obtained for segmentation of brain tumor

TABLE I. EVALUATION METRICS FOR SEGMENTATION MODEL	
Performance metrics	Result
Dice Co-efficient	0.94
F1	0.76
Jaccard	0.68
Recall	0.74
Precision	0.82

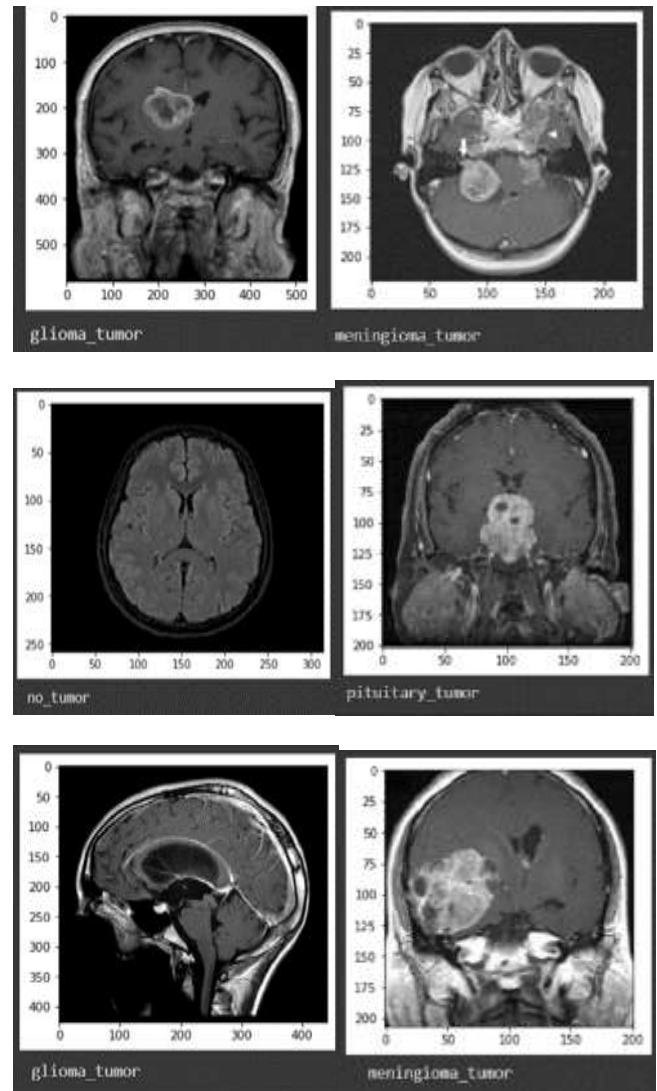


Fig. 6. Result for classification of brain tumor

TABLE II. PERFORMANCE METRICS FOR THE PROPOSED CLASSIFICATION MODEL				
	Precision	Recall	F1-score	Support
Glioma	0.99	0.97	0.98	97
No Tumor	0.96	0.98	0.97	100
Meningioma	1.00	0.96	0.98	49
Pituitary	0.98	1.00	0.99	81

The accuracy plot is shown in Fig 7. The suggested model converges, as seen in the figure, after about 20 epochs, and accuracy of nearly 99% has been attained for both the training and validation data.

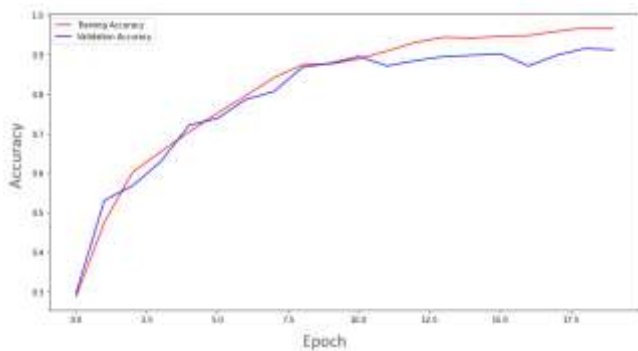


Fig. 7. Accuracy graphs for the VGG16-based classification model

VI. CONCLUSION

Our study aimed to explore the feasibility of using U-Net, a pre-existing deep learning biomedical image system segmentation, in brain tumor segmentation by fine-tuning the model. Our use of U-Net is optimized for accuracy while being lightweight, and doesn't require intensive data augmentation. This demonstrated model can serve as a useful tool for trained physicians to evaluate a patient's MRI images as a second evaluator. CNN model for detecting brain tumors in MRI scans and trained it using a VGG 16, which has 16 layers and is widely used in evaluating commercial models. The objective of our study was to use the VGG 16 CNN model architecture and weights to train the data and detect brain tumors.

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